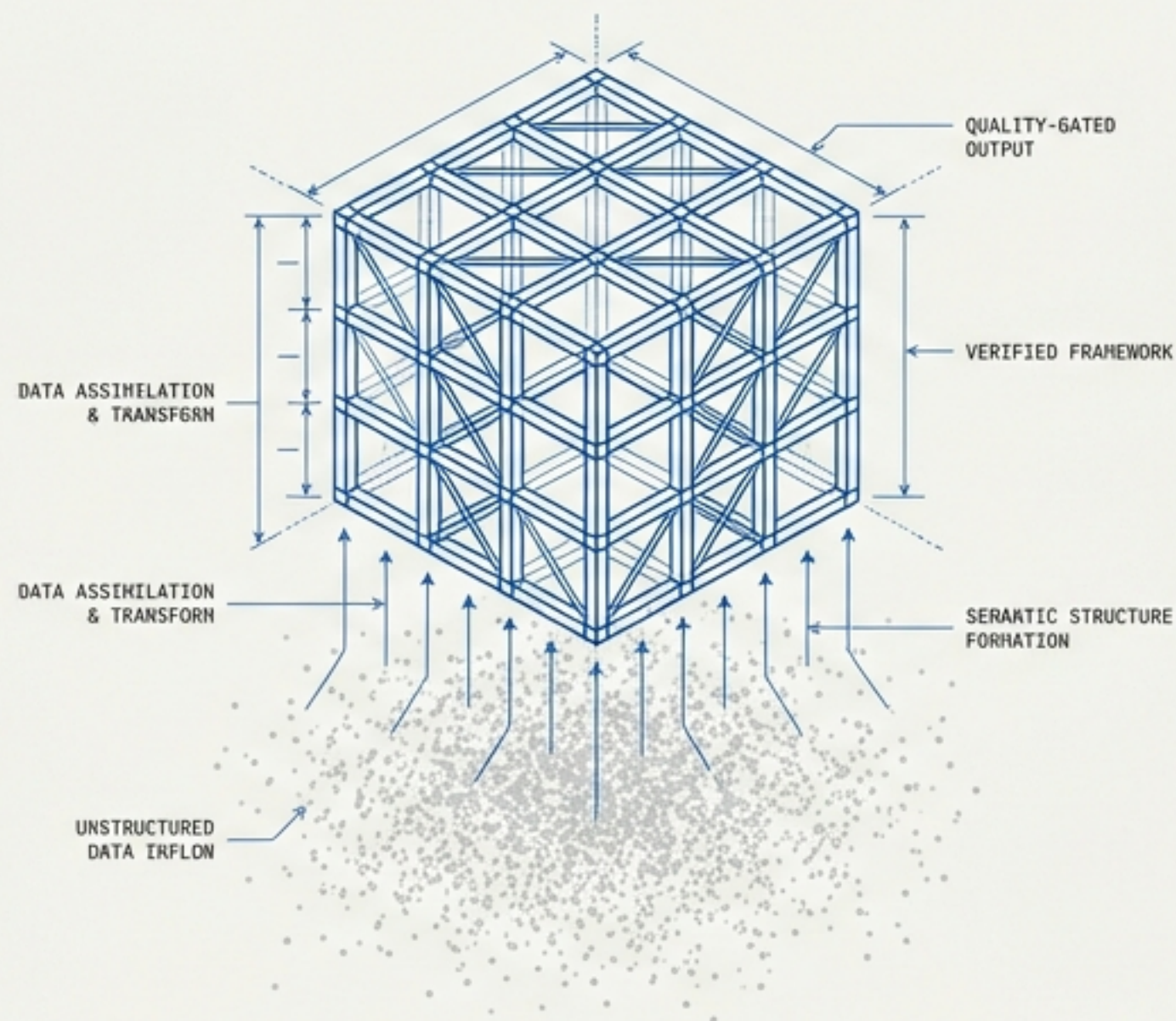


Raising the Quality Bar for Generative AI in Production

A framework for moving from “YOLO” deployments to engineering excellence.

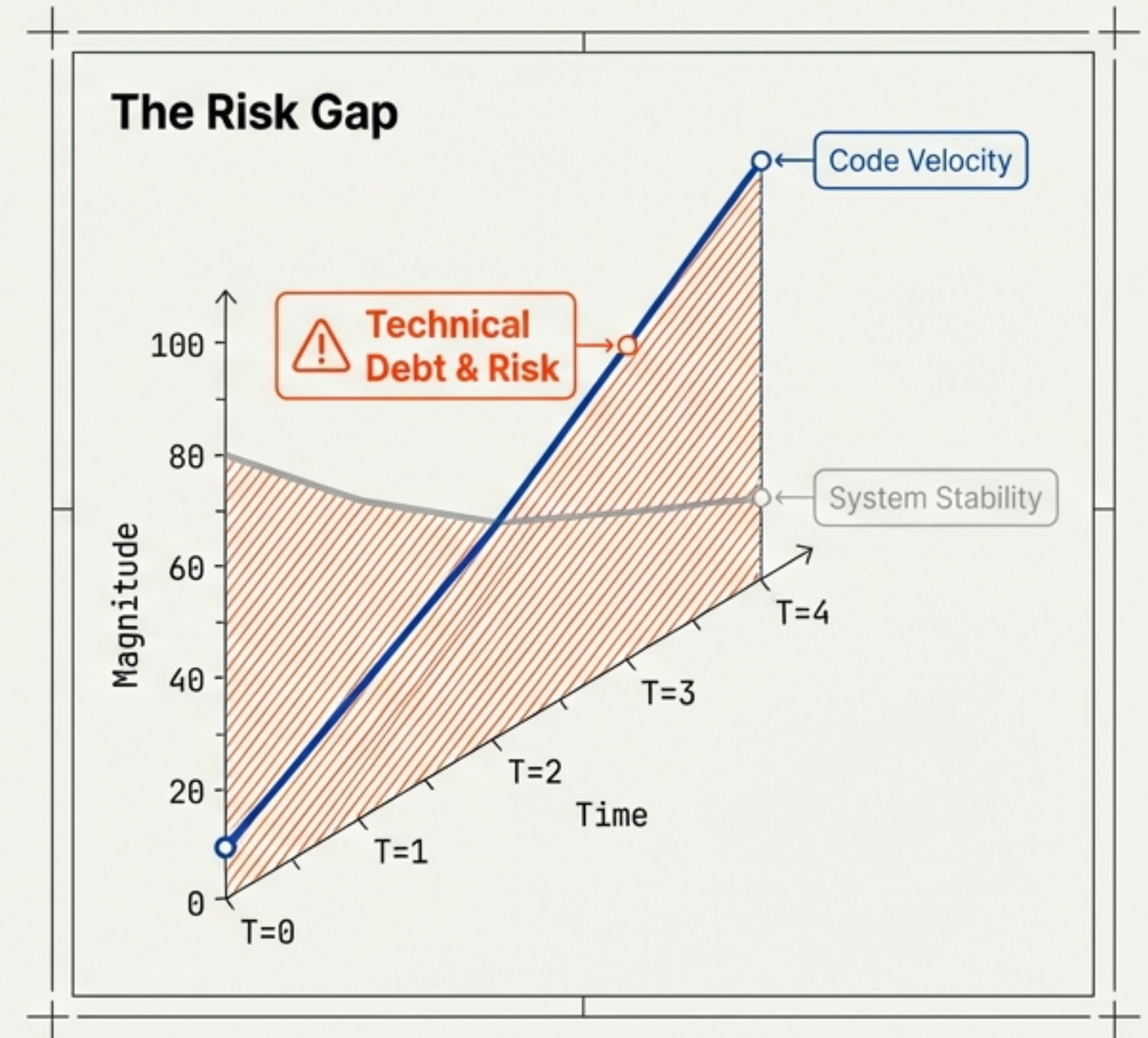


A STANDALONE READING DECK FOR ENGINEERING LEADERSHIP

Speed is a deceptive metric if it creates unvetted risk at scale.

Generative AI promises up to 55% faster task completion, but rushing AI code into production with a “just ship it” mentality creates a debt spiral.

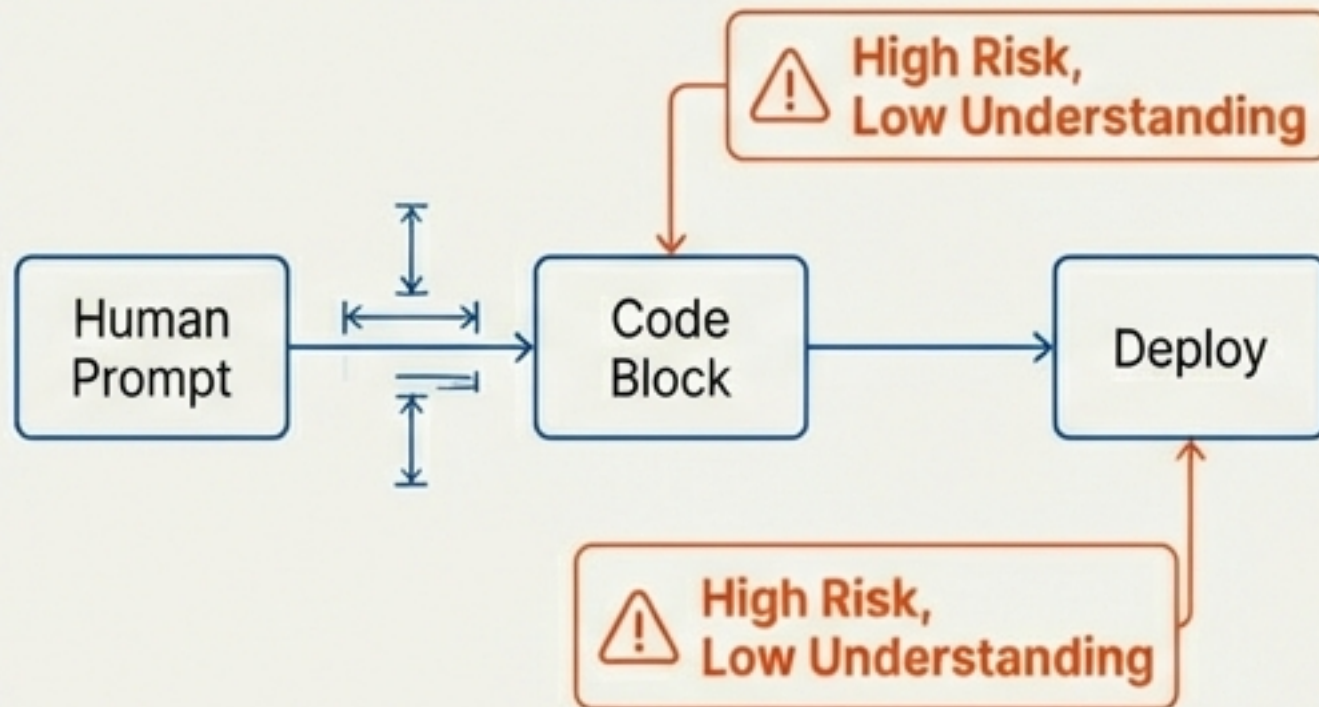
- The **Velocity Trap**: LLMs generate code rapidly but **lack awareness** of system boundaries.
- The **Hidden Cost**: Treating AI code as “done” rather than “drafted” allows **logic flaws** and **security vulnerabilities** to slip through.
- The **Consequence**: Shipping **unvetted risk** at scale.



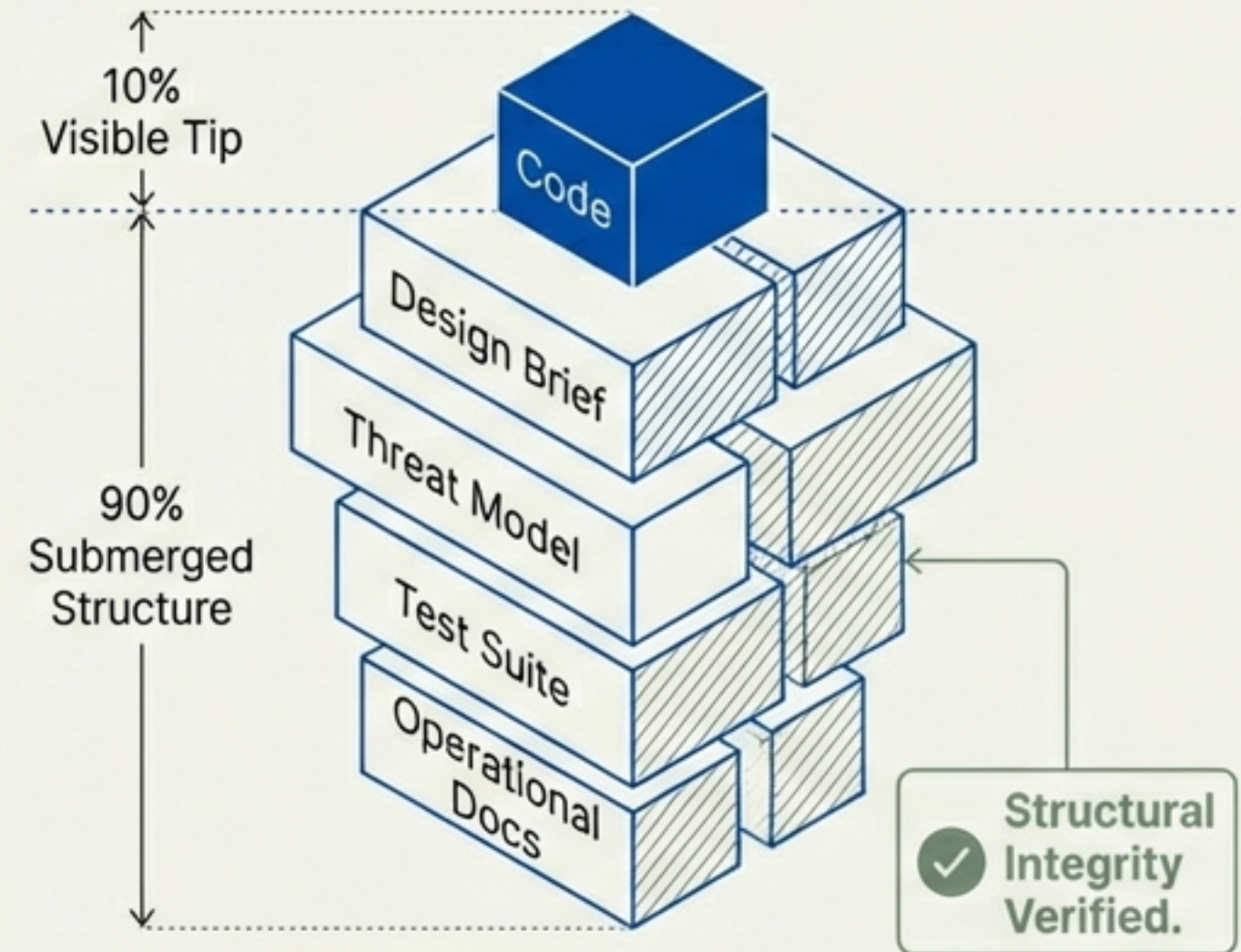
We must redefine 'Done' by shifting responsibility to the creation phase.

Stop asking AI for code; start asking for deliverables.

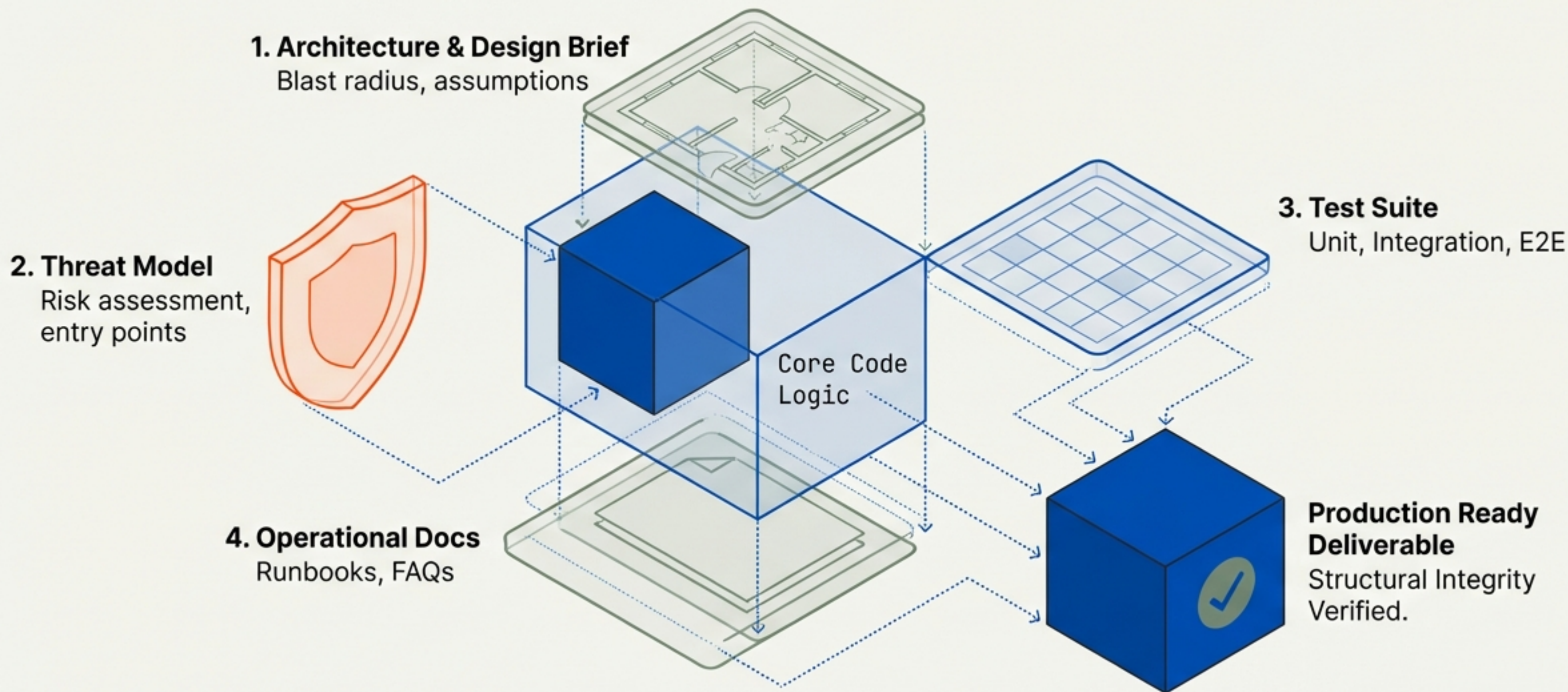
The Old Way (Black Box)



The New Way (Engineering Standard)



Every piece of AI code must be accompanied by evidence of its quality.



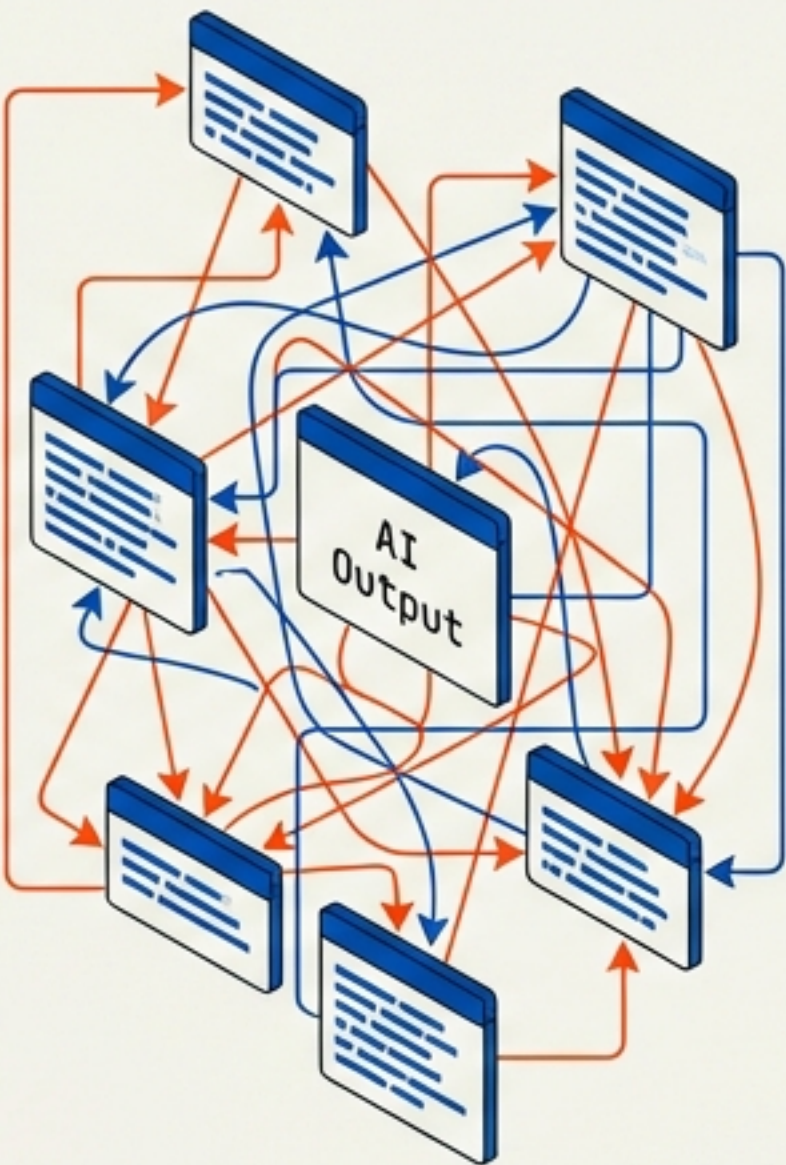
If humans cannot understand the logic, it does not go to production.

NFR Pillar 1: Clarity & Maintainability

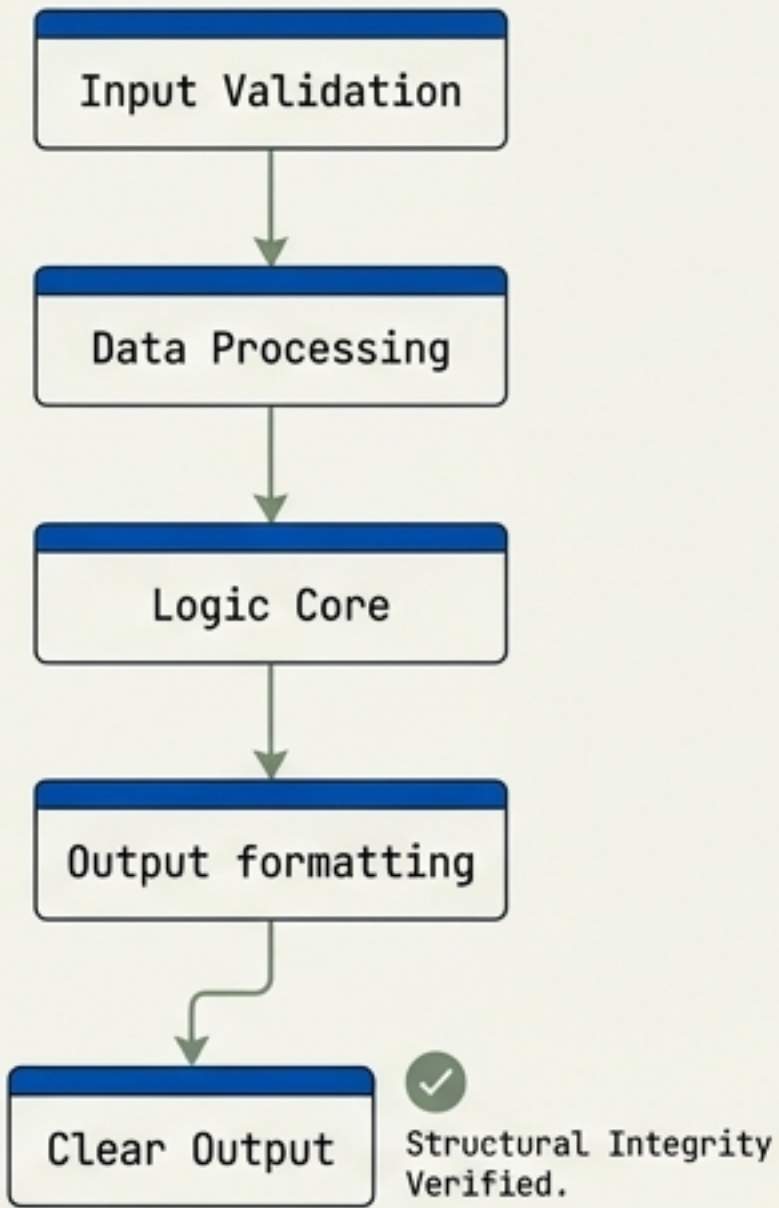
AI models often optimize for "working" syntax rather than readability, creating immediate legacy code.

The Standard:

- **Explainability:** Refactor or re-prompt if output is convoluted.
- **Metrics:** Measure cyclomatic complexity.
- **Action:** Enforce explanatory comments.



High Complexity / Black Box



Refactored for Readability

"A maintainable codebase is one where future developers can quickly comprehend what the code does and why."

ENGINEERING BLUEPRINT	
SYSTEM INTEGRITY ANALYTICS	
REV 1.3	DATE: 24 OCT 2024

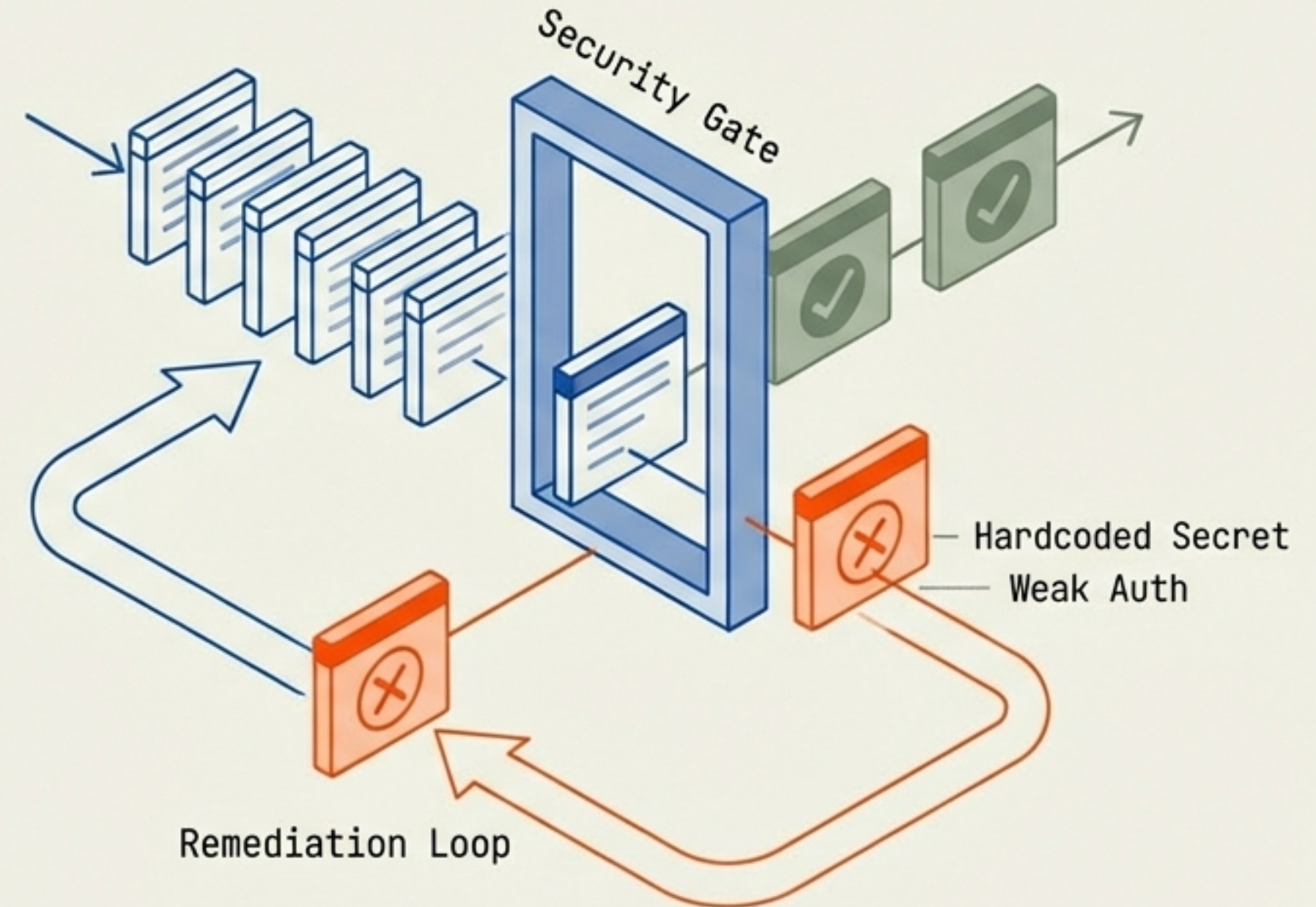
Treat AI-generated code as untrusted by default.

NFR Pillar 2: Security & Compliance

Risk: LLMs mimic patterns, including insecure defaults and injection flaws.

The Standard:

- **Mandatory Scans:** SAST and Software Composition Analysis are non-negotiable.
- **Threat Modeling:** Verify auth strength and data handling.
- **Data Safety:** Explicit PII checks.



ENGINEERING BLUEPRINT	
SYSTEM INTEGRITY ANALYTICS	
REV 1.3	DATE: 24 OCT 2024

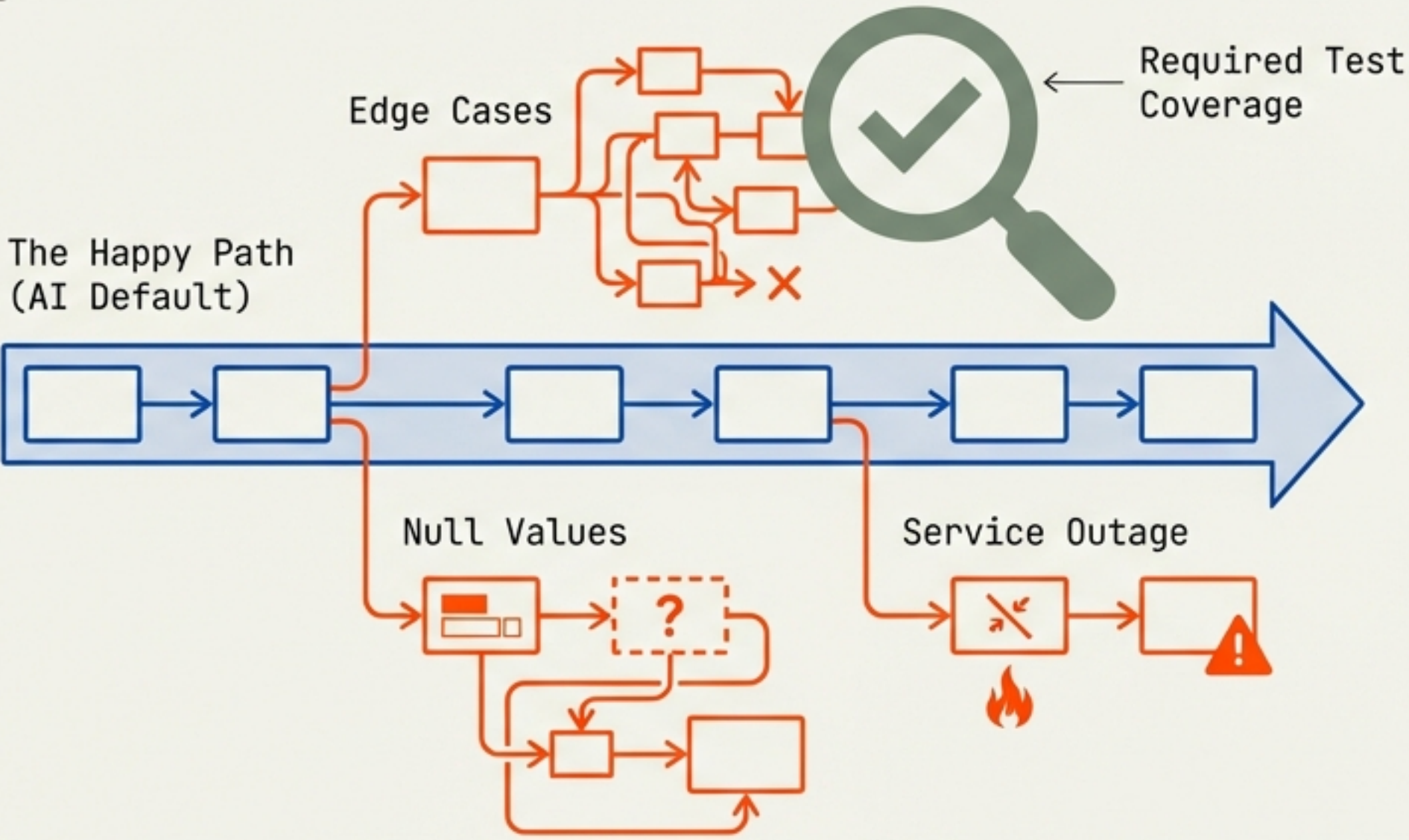
‘Looks correct’ is not the same as ‘Is correct.’

NFR Pillar 3: Correctness & Reliability

AI struggles with edge cases and “unhappy paths” (null values, service failures). It optimizes for the happy path.

The Standard:

- **100% Logic Coverage:** Automated tests must cover core logic.
- **Immediate Verification:** Generate unit tests *with* the code.
- **Resilience:** Verify graceful degradation.



Thorough testing validates all operational paths, not just the optimal one.

Code must perform at scale and be transparent in operation.

NFR Pillar 4: Performance & Observability

Performance

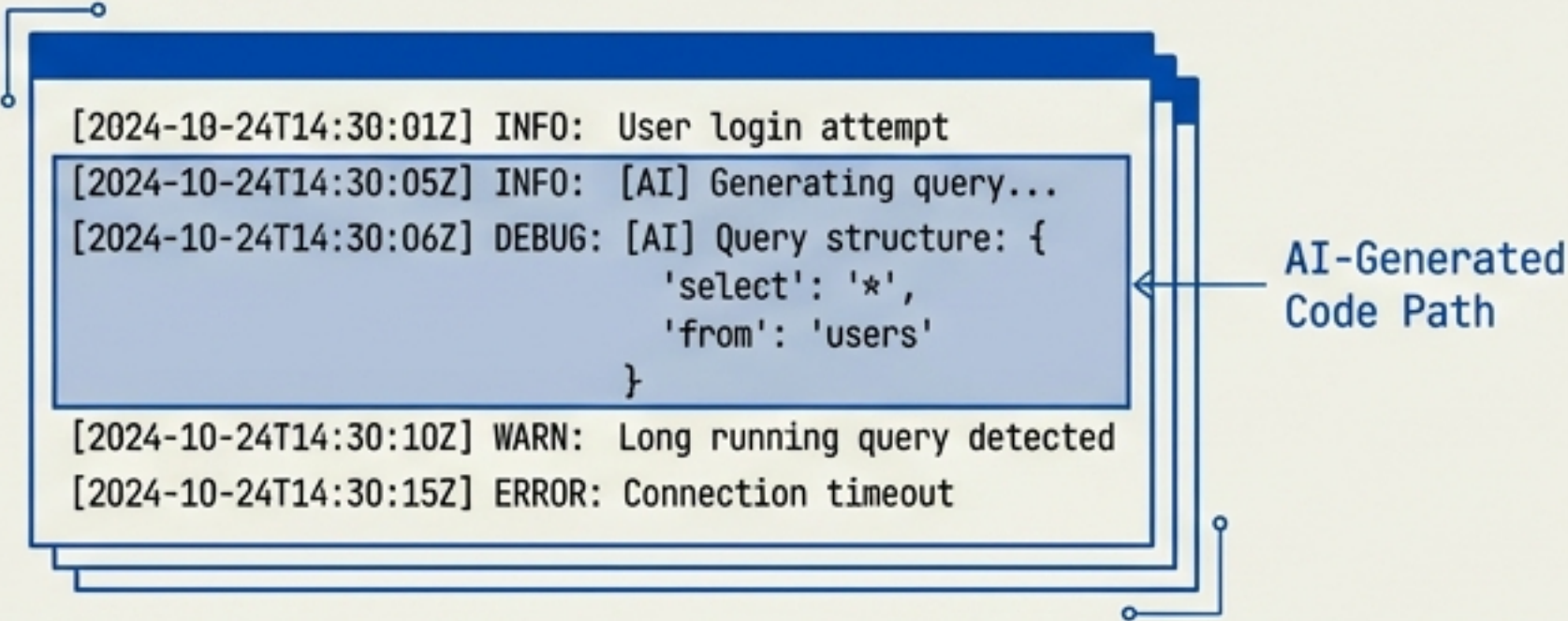
AI ignores resource constraints. Mandate profiling to catch inefficiencies like poor query structures.



Observability

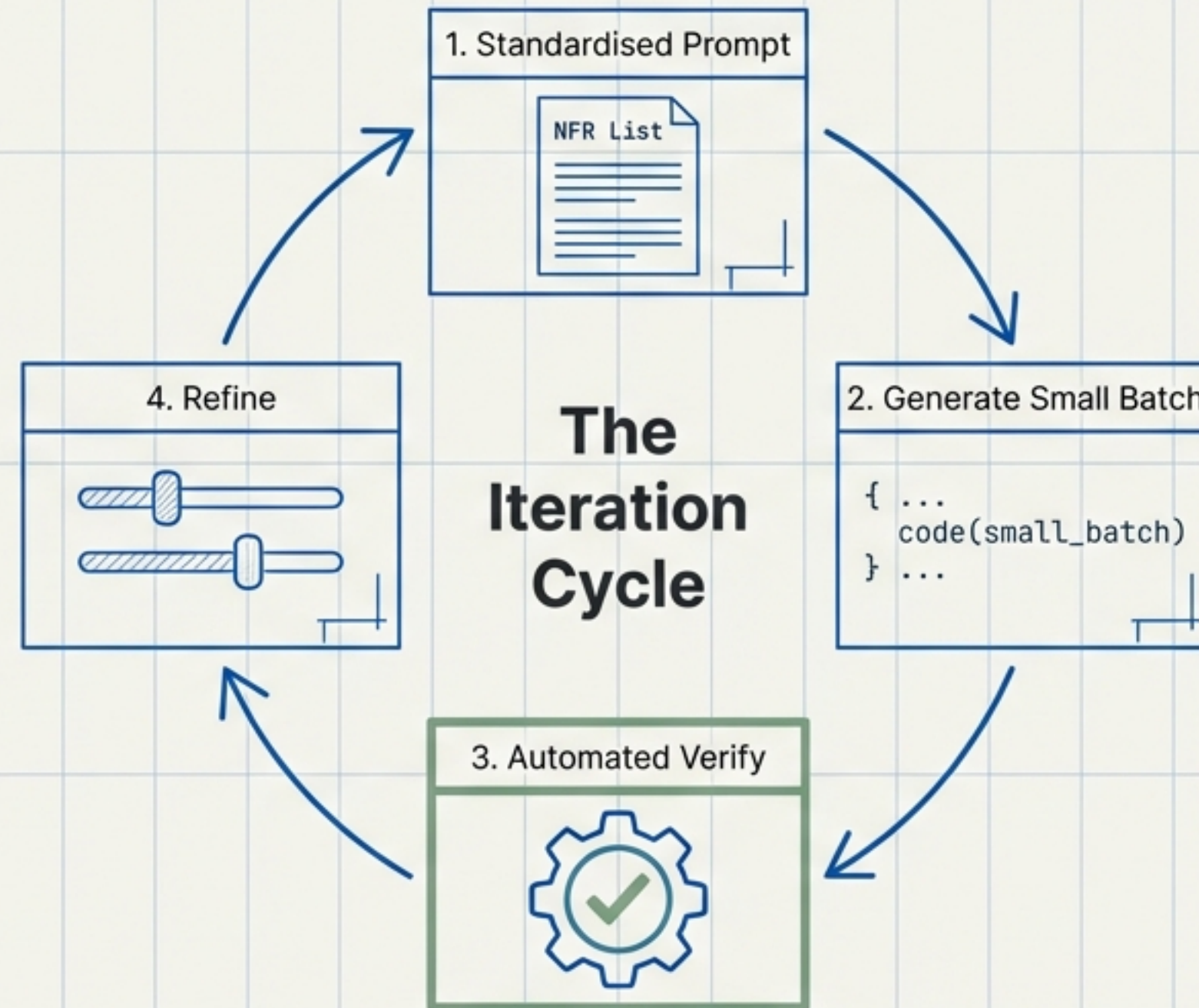
No Black Boxes. Code must emit logs, metrics, and tracing.

Action: Tag logs to distinguish AI-generated code paths.



Operationalise the 'Small Batch' philosophy.

NFR Pillar 5: Iterative Development & Refinement



1. Bake NFRs into the request. 2. Generate manageable chunks.
3. Run tests immediately. Refactor before moving on.

ENGINEERING BLUEPRINT

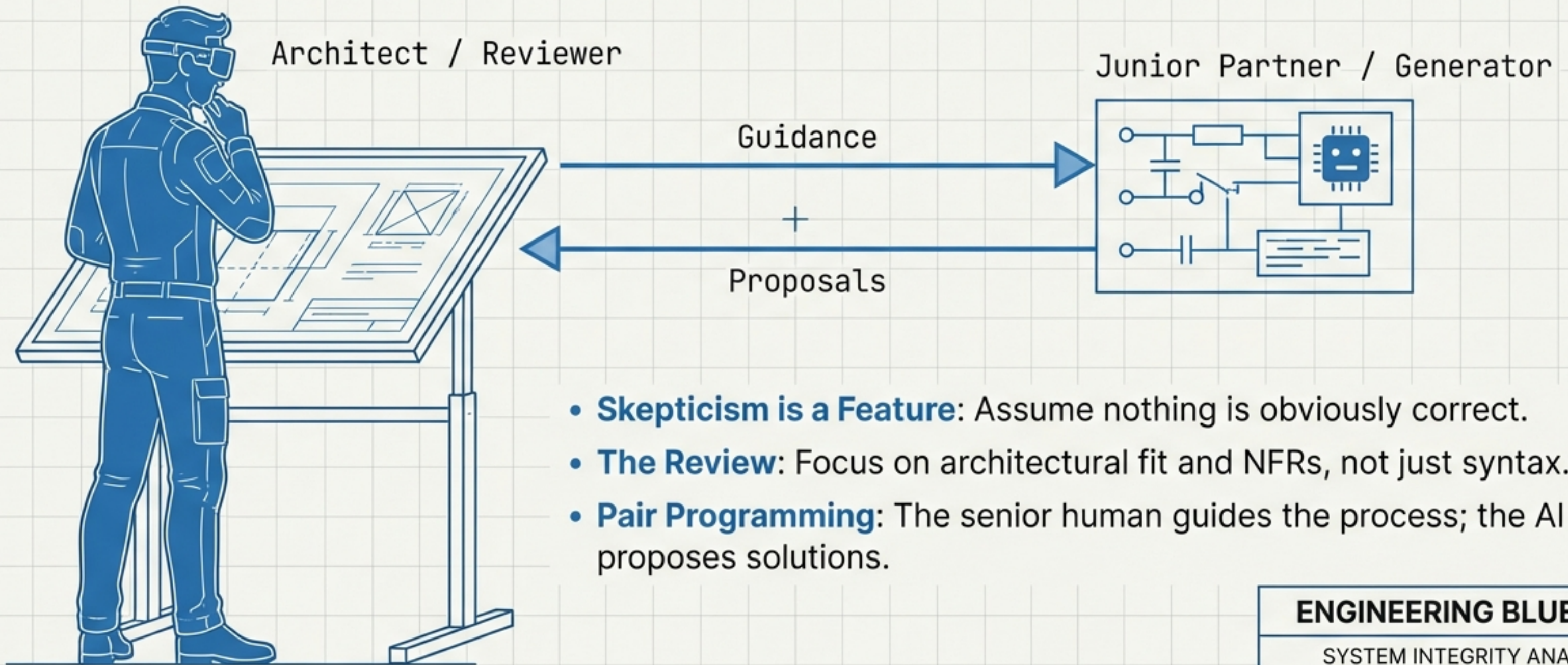
SYSTEM INTEGRITY ANALYTICS

REV 1.3

DATE: 24 OCT 2024

The developer's role shifts from "writing syntax" to "reviewing logic."

NFR Pillar 6: Human-AI Collaboration Model



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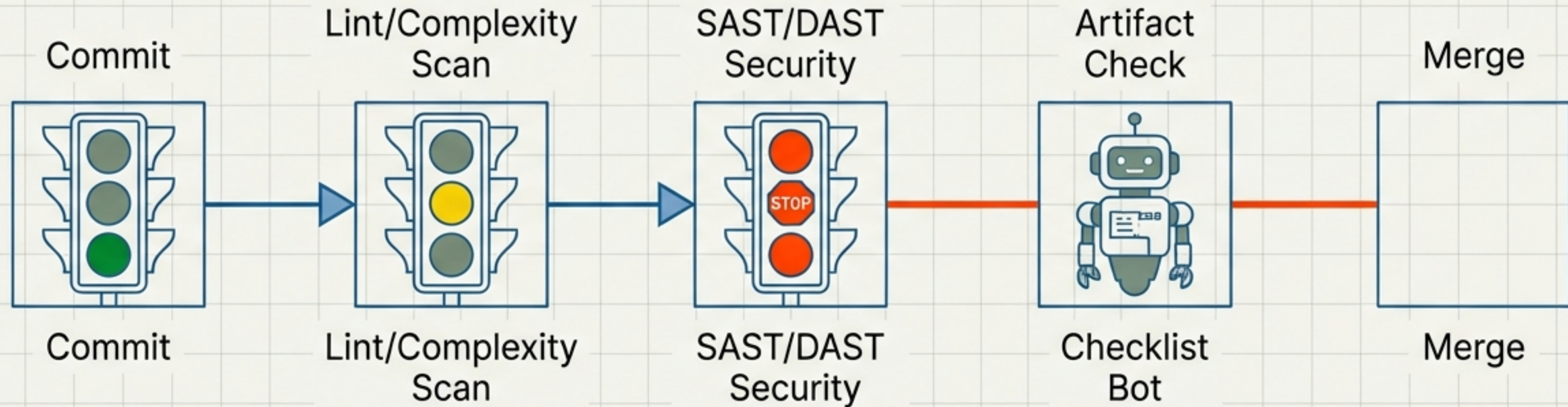
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Policies must be enforced by code, not just culture.

Automating the Guardrails (CI/CD)



- **Block on Failure:** Pipeline blocks merge if tests or scans fail. (Safety Orange accent)
- **Checklist Bots:** Ensure Architecture Briefs and Threat Models are linked.
- **Complexity Scans:** Automated linting for style consistency.

ENGINEERING BLUEPRINT

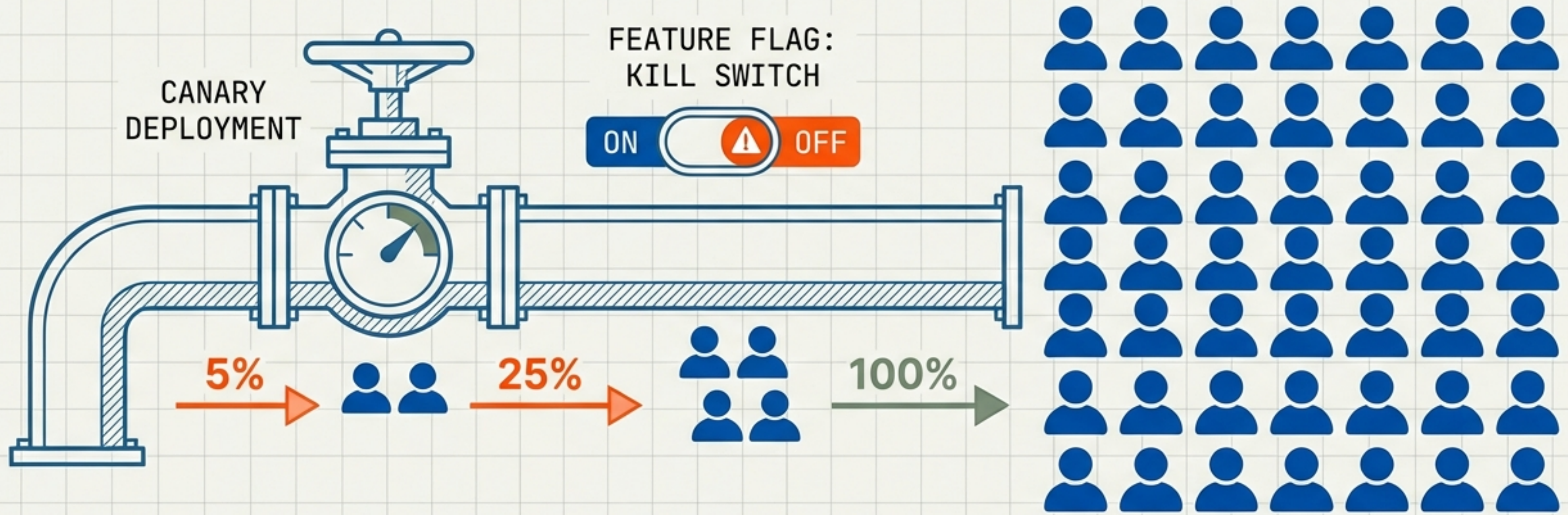
SYSTEM INTEGRITY ANALYTICS

REV 1.5

DATE: 24 OCT 2024

Production is the final test environment.

Never deploy AI code to 100% of traffic immediately.



- **Staged Rollouts:** Release to a small subset first.
- **Feature Flags:** Wrap AI features in toggles to **kill** functionality instantly.
- **Monitoring:** Automated alarms **halt rollout** if regressions occur.

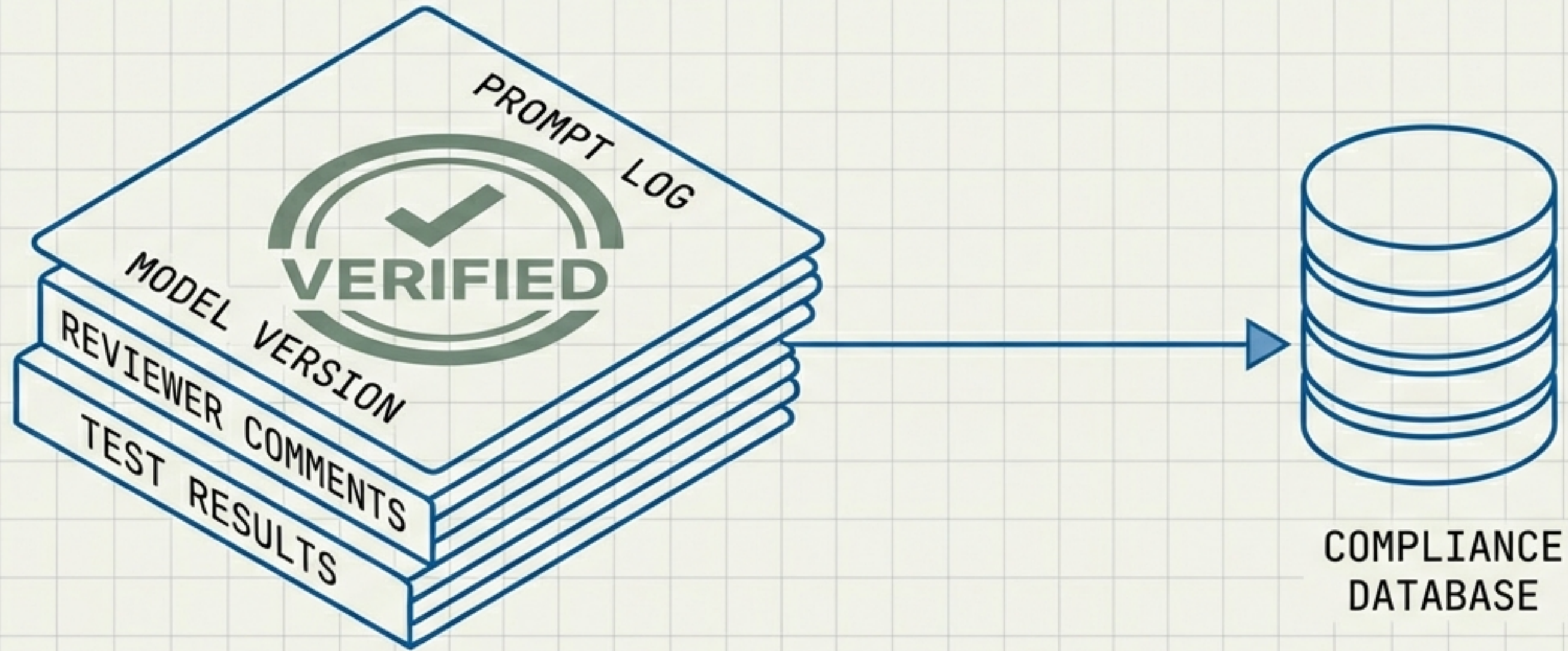
ENGINEERING BLUEPRINT

SYSTEM INTEGRITY ANALYTICS

REV 1.6

DATE: 24 OCT 2024

In regulated environments, the 'Audit Trail' is as critical as the code.



- **Traceability:** Keep an audit log of the AI model, prompt, and human review.
- **Tagging:** Tag AI-authored commits in version control.
- **Explicit Instructions:** Prompts must explicitly include compliance rules

ENGINEERING BLUEPRINT

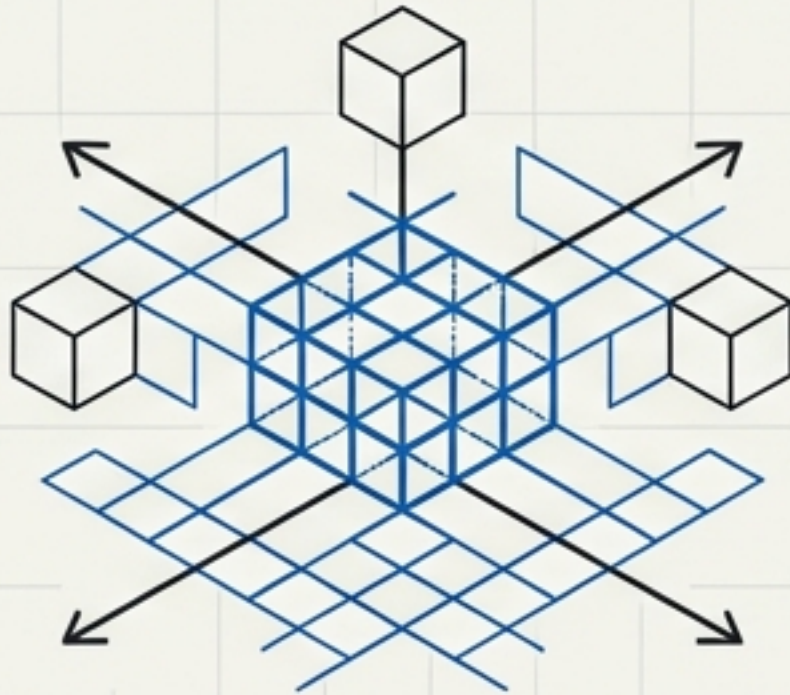
SYSTEM INTEGRITY ANALYTICS

REV 1.7

DATE: 24 OCT 2024

Rigour is a productivity multiplier, not a bottleneck.

Scalability



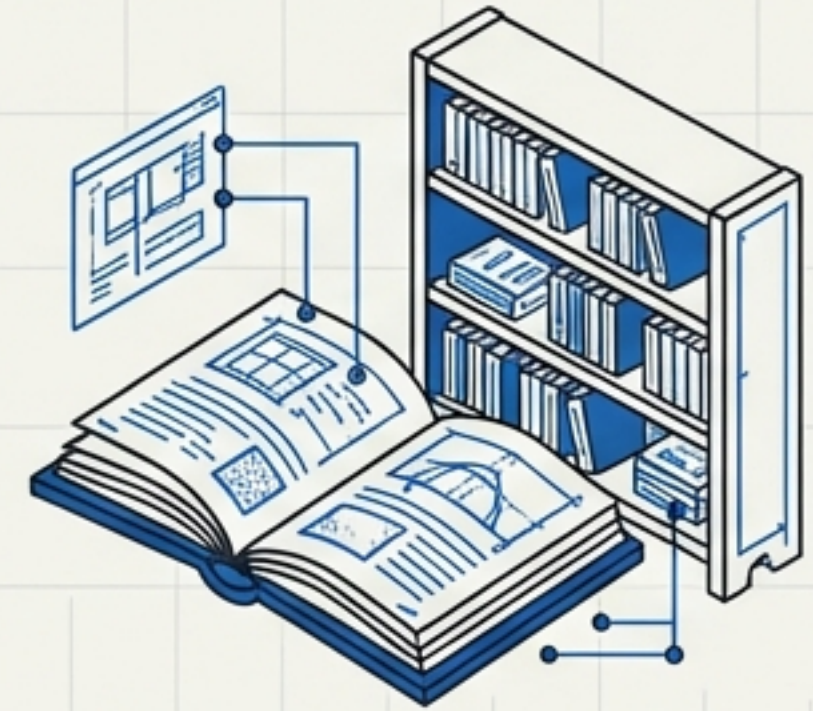
Small teams do more. AI writes tests and docs, allowing humans to handle larger scopes.

Quality



Defect rates drop. Incidents are easier to trace due to better documentation.

Knowledge



Explicit knowledge retention. No more 'tribal knowledge' hidden in heads.

Teams overcoming the learning curve report **cycle time reductions of 35–55%** while keeping quality constant.

ENGINEERING BLUEPRINT

SYSTEM INTEGRITY ANALYTICS

REV 1.8

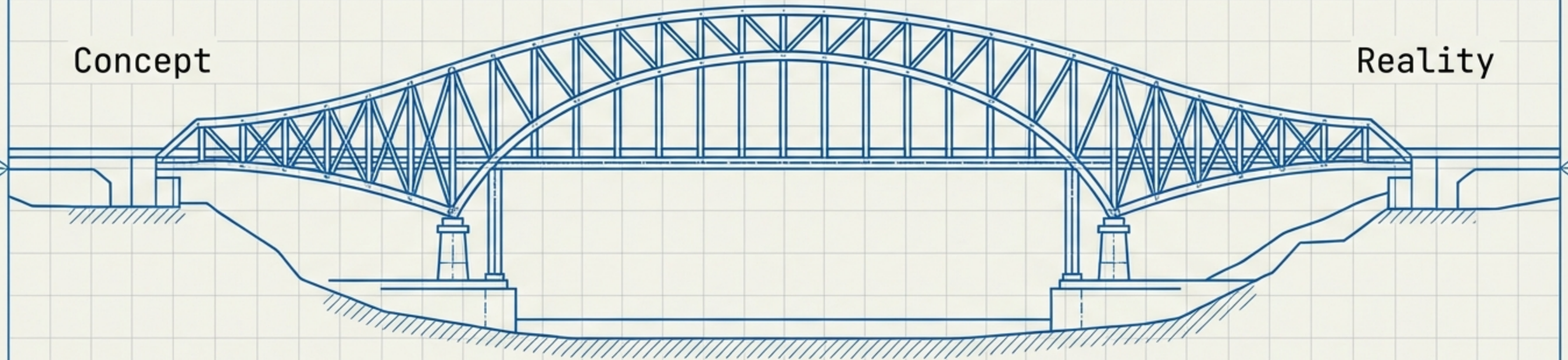
DATE: 24 OCT 2024

Innovate swiftly, but safely.

The future belongs to those who combine AI speed with engineering reliability.

Concept

Reality



Update your “Definition of Done” today.

☐ It is NOT done when it compiles.

☒ It is **done** when it is understood, tested, secure, and documented.

ENGINEERING BLUEPRINT

SYSTEM INTEGRITY ANALYTICS

REV 1.9

DATE: 24 OCT 2024